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Распиновка M.2 (NGFF) connector

разъем винчестера

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Разъем 75 pin M.2 Edge (B and M notched)

M.2 (Next Generation Form Factor, NGFF), is a specification for computer expansion cards. Having a small and flexible physical specification, the M.2 is suitable for solid-state storage applications, especially when used in small devices such as ultrabooks or tablets.

The M.2 standard is designed as a revision and improvement to the mSATA standard, which uses the PCI Express Mini Card physical layout. While mSATA took advantage of the existing PCI Express Mini Card form factor and connector, M.2 has been designed to maximize usage of the card space. Supported host controller interfaces and internally provided ports are a superset to those defined by the SATA Express interface.

The M.2 specification covers multiple Host Interface solutions including:

- PCI Express (PCIe)
- Serial Peripheral Interface (SPI)
- High-Speed Inter-Chip (HSIC)
- SuperSpeed Inter-Chip (SSIC)
- Mobile PCIe (M-PCIe)
- Universal Serial Bus (USB)
- Secure Digital Input Output (SDIO)
- Universal Asynchronous Receiver/Transmitter (UART)
- Pulse Code Modulation / Inter-IC Sound (PCM/I2S)
- Inter-Integrated Circuit (I2C)
- System Management Bus (SMBus)
- Serial ATA (SATA)
- DisplayPort (DP)
- All future variants of the interfaces in this list

Computer bus interfaces provided through the M.2 connector are PCI Express 3.0 (up to four lanes), Serial ATA 3.0, and USB 3.0 (a single logical port for each of the latter two). It is up to the manufacturer of the M.2 host or device to select which interfaces are to be supported, depending on the desired level of host support and device type. The M.2 connector has different keying notches that denote various purposes and capabilities of M.2 hosts and modules, preventing plugging of M.2 modules into feature-incompatible host connectors

M.2 uses a dual-sided edge card connector with a 0.5 mm contact pitch. There are some mechanical keys defined for SSDs. For example key B M.2 pinout supports SSD/WWAN: 1x SATA SSD or 1x, 2x PCIe SSD (and WWAN) Host Interfaces. Key M M.2 pinout supports SSDs only: 1x SATA or 1x, 2x, or 4x PCIe Host Interfaces. Sockets are defined as follows: Socket 1 accepts cards with an "A" key notch present, Socket 2 accepts cards with a "B" key notch present, Socket 3 accepts cards with an "M" key notch present.

M.2 pinout for key B (1x SATA, 2x PCIe)

Pin Number	Pin Name	Description	Voltage
1	CONFIG_3	Defines module type	
2	3.3 V	Supply pin	3.3 V
3	GND	Ground	
4	3.3 V	Supply pin	3.3 V
5	GND		

Статус распиновки: +5 -0

На распиновку M.2 (NGFF) connector есть 8 отзыв(а): 5 положительных и 0 отрицательных.

Эта распиновка правильная с ошибками

Нет DIY устройств

2 источников информации

- M2 schematic
- wikipedia

6	FULL_CARD_POWER_OFF#	Out	0/1.8V or 3.3V
7	USB_D+		
8	W_DISABLE#	Out	0/3.3V
9	USB D-		
10	GPIO_9/DAS/DSS	Device Activity Signal / Disable Staggered Spinup	0/3.3V
11	GND		
12-19	removed	Mechanical notch B	
20	GPIO_5	I/O	0/1.8V
21	CONFIG_0	Defines module type	
22	GPIO_6	I/O	0/1.8V
23	GPIO_11	I/O	0/1.8V
24	GPIO_7	I/O	0/1.8V
25	DPR	Out	0/1.8V
26	GPIO_10	I/O	0/1.8V
27	GND	Ground	
28	GPIO_8	I/O	0/1.8V
29	PERn1 / USB RX- /SSIC-RxN	PCIe Lane 1 Rx or USB 3.0 RX-	
30	UIM-RESET	In	
31	PERp1 / USB RX+/ SSIC-RxP	PCIe Lane 1 Rx or USB 3.0 RX+	
32	UIM-CLK	In	
33	GND	Ground	
34	UIM-DATA	I/O	
35	PETn1 / USB TX- /SSIC-TxN	PCIe Lane 1 Tx or USB 3.0 TX-	
36	UIM-PWR	In	
37	PETp1 / USB TX+/SSIC-TxP	PCIe Lane 1 Tx or USB 3.0 TX+	
38	DEVSLP	Device Sleep, input. If driven high the host is informing the SSD to enter a low power state.	
39	GND	Ground	
40	GPIO_0/SMB_CLK	I/O	0/1.8V
41	SATA-B+/PERn0	Host receiver differential	

		signal pair. If in PCIe mode PCIe Lane 0 Rx	
42	GPIO_1/SMB_DATA	I/O	0/1.8V
43	SATA-B-/PERp0	Host receiver differential signal pair. If in PCIe mode PCIe Lane 0 Rx	
44	GPIO_2/ALERT#		0/1.8V
45	GND	Ground	
46	GPIO_3	I/O	0/1.8V
47	SATA-A-/PETn0	Host transmitter differential signal pair. If in PCIe mode PCIe Lane 0 Tx	
48	GPIO_4	I/O	0/1.8V
49	SATA-A+/PETp0	Host transmitter differential signal pair. If in PCIe mode PCIe Lane 0 Tx	
50	PERST#	PCIe reset	0/3.3V
51	GND	Ground	
52	CLKREQ#	Reference clock request signal	0/3.3V
53	REFCLKN	PCIe Reference Clock signals (100 MHz)	
54	PEWAKE#	PCIe WAKE# Open Drain with pull up on platform. Active Low.	0/3.3V
55	REFCLKP	PCIe Reference Clock signals (100 MHz)	
56	MFG1	Manufacturing pin. Use determined by vendor.	
57	GND	Ground	
58	MFG2	Manufacturing pin. Use determined by vendor.	
59	ANTCTL0	In	0/1.8V
60	COEX3	In	0/1.8V
61	ANTCTL1	In	0/1.8V
62	COEX_TXD	In	0/1.8V

63	ANTCTL2	In	0/1.8V
64	COEX_RXD	In	0/1.8V
65	ANTCTL3	In	0/1.8V
66	SIM DETECT	Out	
67	RESET#	Out	0/1.8V
68	SUSCLK	32.768 kHz clock supply input provided by the Platform chipset	0/3.3V
69	CONFIG_1	Defines module type	
70	3.3V / VBAT	Supply pin	3.3V
71	GND	Ground	
72	3.3V / VBAT	Supply pin,	3.3V
73	GND	Ground	
74	3.3V / VBAT	Supply pin	3.3V
75	CONFIG_2	Defines module type	

CONFIG pins are set by the SSD to inform the host if the drive wishes to use the SATA or PCIe signaling scheme. SSD-SATA pull pin 1, pin 21, pin 69, pin 75 to Ground.
Config pins configuration:

Config_0 pin 21	Config_1 pin 69	Config_2 pin 75	Config_3 pin 1	Host Interface
0	0	0	0	SSD-SATA
0	1	0	0	SSD-PCIe
0	0	1	0	WWAN – PCIe (Port Configuration 0)
0	1	1	0	WWAN – PCIe (Port Configuration 1)
0	0	0	1	WWAN – PCIe, USB3.1 Gen1 (Port Configuration 0)
0	1	0	1	WWAN – PCIe, USB3.1 Gen1 (Port Configuration 1)
0	0	1	1	WWAN – PCIe, USB3.1 Gen1 (Port Configuration 2)
0	1	1	1	WWAN – PCIe, USB3.1 Gen1 (Port Configuration 3)
1	0	0	0	WWAN – SSIC (Port Configuration 0)
1	1	0	0	WWAN – SSIC (Port Configuration 1)

Config_0 pin 21	Config_1 pin 69	Config_2 pin 75	Config_3 pin 1	Host Interface
1	0	1	0	WWAN – SSIC (Port Configuration 2)
1	1	1	0	WWAN – SSIC (Port Configuration 3)
1	0	0	1	WWAN – PCIe (Port Configuration 2)
1	1	0	1	WWAN – PCIe (Port Configuration 3)
1	0	1	1	WWAN – PCIe, USB3.1 (vendor defined)
1	1	1	1	No Add-in card present

M.2 pinout for key M (1x SATA, 1x, 2x, or 4x PCIe)

Pin Number	Pin Name	Description	Voltage
1	CONFIG_3	Defines module type	
2	3.3 V	Supply pin	3.3 V
3	GND	Ground	
4	3.3 V	Supply pin	3.3 V
5	PERn3	PCIe Lane 3 Rx	
6	N/A		
7	PERp3	PCIe Lane 3 Rx	
8	N/A		
9	GND	Ground	
10	DAS/DSS	Device Activity Signal / Disable Staggered Spinup	
11	PETn3	PCIe Lane 3 Tx	
12	3.3 V	Supply pin	3.3 V
13	PETp3	PCIe Lane 3 Tx	
14	3.3 V	Supply pin	3.3 V
15	GND	Ground	
16	3.3V	Supply pin	3.3 V
17	PERn2	PCIe Lane 2 Rx	
18	3.3 V	Supply pin	3.3 V
19	PERp2	PCIe Lane 2 Rx	
20	N/A		
21	CONFIG_0	Defines module type	
22	N/A		
23	PETn2	PCIe Lane 2 Tx	

24	N/A		
25	PETp2	PCIe Lane 2 Tx	
26	N/A		
27	GND	Ground	
28	N/A		
29	PERn1	PCIe Lane 1 Rx	
30	N/A		
31	PERp1	PCIe Lane 1 Rx	
32	N/A		
33	GND	Ground	
34	N/A		
35	PETn1	PCIe Lane 1 Tx	
36	N/A		
37	PETp1	PCIe Lane 1 Tx	
38	DEVSLP	Device Sleep, input. If driven high the host is informing the SSD to enter a low power state.	
39	GND	Ground	
40	N/A		
41	SATA-B+/PERn0	Host receiver differential signal pair. If in PCIe mode PCIe Lane 0 Rx	
42	N/A		
43	SATA-B-/PERp0	Host receiver differential signal pair. If in PCIe mode PCIe Lane 0 Rx	
44	N/A		
45	GND	Ground	
46	N/A		
47	SATA-A-/PETn0	Host transmitter differential signal pair. If in PCIe mode PCIe Lane 0 Tx	
48	N/A		
49	SATA-A+/PETp0	Host transmitter differential signal pair. If in PCIe mode PCIe Lane 0 Tx	
50	PERST#	PCIe reset	
51	GND	Ground	
52	CLKREQ#	Reference clock request signal	
53	REFCLKN	PCIe Reference Clock signals (100 MHz)	
54	PEWAKE#	PCIe WAKE# Open Drain with pull up on platform. Active Low.	
55	REFCLKP	PCIe Reference Clock	

		signals (100 MHz)	
56	MFG1	Manufacturing pin. Use determined by vendor.	
57	GND	Ground	
58	MFG2	Manufacturing pin. Use determined by vendor.	
59-66	removed	Mechanical notch M	
67	N/A		
68	SUSCLK	32.768 kHz clock supply input provided by the Platform chipset	
69	CONFIG_1	Defines module type	
70	3.3 V	Supply pin	3.3 V
71	GND	Ground	
72	3.3 V	Supply pin	3.3 V
73	GND	Ground	
74	3.3 V	Supply pin	3.3 V
75	CONFIG_2	Defines module type	

M.2 dual module key A and E

Pin id.	Pin name	Description	Voltage
1	GND	Ground	
2	+3.3V	power supply	3.3 V
3	USB_D+	USB high-, full-, and low-speed data pair positive	
4	+3.3V	power supply	3.3 V
5	USB_D-	USB high-, full-, and low-speed data pair negative	
6	LED1#		
7	GND	Ground	
8-15	Key	Substrate removed to act as physical key	
16	LED2#		
17	DNC	Do not connect	
18	GND	Ground	
19	DNC	Do not connect	
20	DNC	Do not connect	
21	DNC	Do not connect	
22	DNC	Do not connect	
23-31	Key	Substrate removed to act as physical key	
32	DNC	Do not connect	
33	GND	Ground	
34	DNC	Do not connect	

35	PETp0	PCI Express lane 0 module transmitter pair positive	
36	DNC	Do not connect	
37	PETn0	PCI Express lane 0 module transmitter pair negative	
38	Vendor defined		
39	GND	Ground	
40	Vendor defined		
41	PERp0	PCI Express lane 0 module receiver pair positive	
42	Vendor defined		
43	PERn0	PCI Express lane 0 module receiver pair negative	
44	COEX3	Antenna coexistence signal 3	
45	GND	Ground	
46	COEX2	Antenna coexistence signal 2	
47	PEFCLKP0	PCI Express reference clock pair positive	
48	COEX1	Antenna coexistence signal 1	
49	PEFCLKN0	PCI Express reference clock pair negative	
50	SUSCLK	32.768 kHz clock module input	
51	GND	Ground	
52	PERST0#	PCI Express reset	
53	CLKREQ0#	PCI Express clock request	
54	W_DISABLE2#	Wireless disable 2	
55	PEWake0#	PCI Express wake	
56	W_DISABLE1#	Wireless disable 1	
57	GND	Ground	
58	SMB_DATA	SMBus data signal	
59	Reserved		
60	SMB_CLK	SMBus clock signal	
61	Reserved		
62	ALERT#	SMBus alert signal	
63	GND	Ground	
64	Reserved		
65	Reserved		
66	UIM_SWP		
67	Reserved		
68	UIM_POWER_SNK		

69	GND	Ground	
70	UIM_POWER_SRC		
71	Reserved		
72	+3.3V	power supply	3.3 V
73	Reserved		
74	+3.3V	power supply	3.3 V
75	GND	Ground	

M.2 module key E

Pin id.	Pin name	Description	Voltage
1	GND	Ground	
2	+3.3V	3.3 V power supply	
3	USB_D+	USB high-, full-, and low-speed data pair positive	
4	+3.3V	3.3 V power supply	
5	USB_D-	USB high-, full-, and low-speed data pair negative	
6	LED1#		
7	GND	Ground	
8	PCM_CLK/I2S SCK	I/O	0/1.8V
9	SDIO CLK/SYSCLK	Out	0/1.8V
10	PCM SYNC/I2S WS	I/O	0/1.8V
11	SDIO CMD	I/O	0/1.8V
12	PCM_IN/I2S_IN	In	0/1.8V
13	SDIO DATA0	I/O	0/1.8V
14	PCM_OUT/I2S SD_OUT	Out	0/1.8V
15	SDIO DATA1		0/1.8V
16	LED2#	In	
17	SDIO DATA2	I/O	0/1.8V
18	GND	Ground	
19	SDIO DATA3	I/O	0/1.8V
20	UART WAKE#	In	0/3.3V
21	SDIO WAKE#	In	0/1.8V
22	UART RxD	In	0/1.8V
23	SDIO RESET#/TX_BLANKING	Out	0/1.8V
24-31	Key E	Substrate removed to act as physical key	

32	UART TxD	Out	0/1.8V
33	GND	Ground	
34	UART CTS	In	0/1.8V
35	PETp0	PCI Express lane 0 module transmitter pair positive	
36	UART RTS	Out	0/1.8V
37	PETn0	PCI Express lane 0 module transmitter pair negative	
38	Vendor defined		
39	GND	Ground	
40	Vendor defined		
41	PERp0	PCI Express lane 0 module receiver pair positive	
42	Vendor defined		
43	PERn0	PCI Express lane 0 module receiver pair negative	
44	COEX3	Antenna coexistence signal 3	0/1.8V
45	GND	Ground	
46	COEX_RxD	In	0/1.8V
47	PEFCLKP0	PCI Express reference clock pair positive	
48	COEX_TxD	Out	0/1.8V
49	PEFCLKN0	PCI Express reference clock pair negative	
50	SUSCLK	32.768 kHz clock module input	0/3.3V
51	GND	Ground	
52	PERST0#	PCI Express reset	0/3.3V
53	CLKREQ0#	PCI Express clock request	0/3.3V
54	W_DISABLE2#	Wireless disable 2	0/3.3V
55	PEWake0#	PCI Express wake	0/3.3V
56	W_DISABLE1#	Wireless disable 1	0/3.3V
57	GND	Ground	
58	I2C_DATA	I2c data signal	0/1.8V
59	Reserved/PETp1		

60	I2c_CLK	I2C clock signal	0/1.8V
61	Reserved/PETn1		
62	ALERT#	I2c Bus alert signal In	0/1.8V
63	GND	Ground	
64	Reserved		
65	Reserved/PERp1		
66	UIM_SWP/PERST1#		
67	Reserved/PERn1		
68	UIM_POWER_SNK/CLKREQ1#		
69	GND	Ground	
70	UIM_POWER_SRC/PEWAKE1#		
71	Reserved/REFCLKp1		
72	+3.3V	power supply	3.3 V
73	Reserved		
74	+3.3V	power supply	3.3 V
75	GND	Ground	

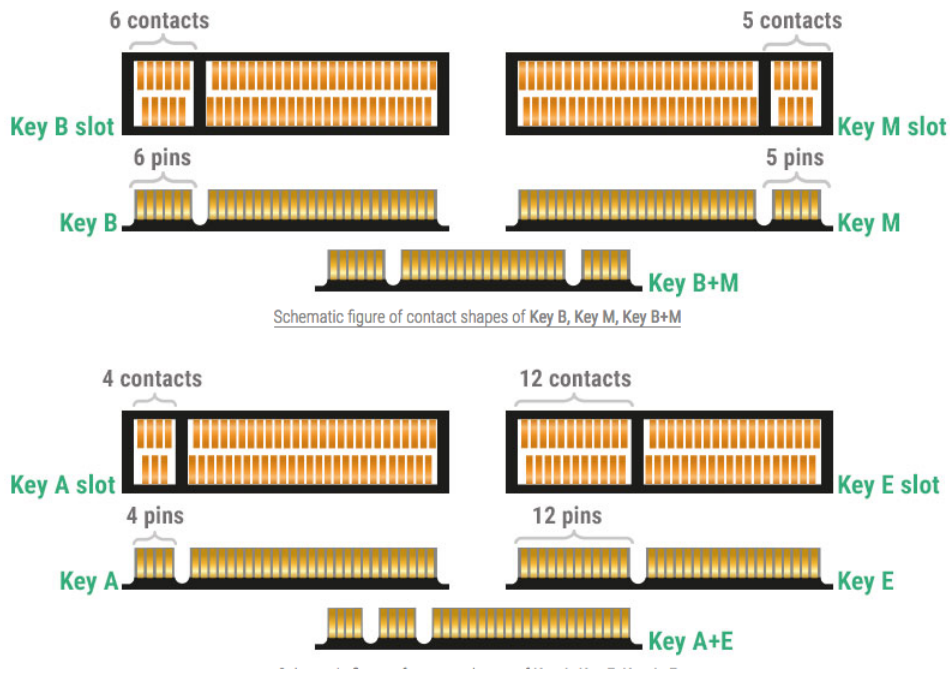
M.2 module keying

Key ID	Pin Location	Interface
A	8-15	2x PCIe x1 / USB 2.0 / I2C /DP x4
B	12-19	PCIe x2 / SATA /USB 2.0 / USB 3.0 / HSIC / SSIC / Audio / UIM / I2C
C	16-23	Reserved for Future Use
D	20-27	Reserved for Future Use
E	24-31	2x PCIe x1 / USB 2.0 / I2C / SDIO / UART / PCM
F	28-35	Future Memory Interface (FMI)
G	39-46	Not Used for M.2; for Custom/Non-Standard Apps
H	43-50	Reserved for Future Use
J	47-54	Reserved for Future Use
K	51-58	Reserved for Future Use
L	55-62	Reserved for Future Use

M	59-66	PCIe x4 / SATA

M.2 SATA SSDs are B+M-keyed (can fit in sockets for B-keyed and M-keyed modules), while M.2 NVMe SSDs for PCIe 3.0 x4 lane are M-keyed.

M.2 key	M.2 module size	M.2 devices
A	1630, 2230, 3030	WiFi, Bluetooth, NFC and/or WiGig
E	1630, 2230, 3030	WiFi, Bluetooth, NFC and/or GPS/GNSS
B	3042, 2230, 2242, 2260, 2280, 22110	3G/LTE + GPS/GNSS or SSD
M	2242, 2260, 2280, 22110	SSD with PCIe or SATA interface



Источник(и) этой и дополнительной информации: Serial ATA Revision 3.2 Gold

Прислал: , M
Последнее обновление 2021-11-20 16:06:24.

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